

Continuum Mechanics

9/17/2026

class # 10

Chapter 3: Continuum Mass and Force

Mass Density

Definition A region $B \subseteq \mathbb{E}^3$ is called regular iff

- 1) B is the union of a finite number of open disjoint and bounded components
- 2) The bounding surface of B , denoted ∂B , is piecewise smooth and consists of a finite number of disjoint components
- 3) Each component of ∂B is orientable, in the sense that it has two distinct sides.

Remark: A regular region can have a finite number of holes.

Definition: Suppose that B is a regular region and $\Omega \subseteq B$ is arbitrary. Assume that there is a scalar field $\rho: B \rightarrow \mathbb{R}$, called the mass density that is smooth and satisfies

$$\rho(\vec{x}) = 0 \quad \forall \vec{x} \in B.$$

Then

$$\text{mass}[\Omega] = \int_{\Omega} \rho(\vec{x}) dV_{\vec{x}}$$

gives the mass of the set Ω .

The volume of the set Ω is denoted

$$\text{vol}[\Omega] = \int_{\Omega} dV_{\mathbf{x}}. \quad //$$

Typically, the units of the mass density field are

$$[\rho] = \frac{\text{kg}}{\text{m}^3}$$

and

$$[\text{mass}[\Omega]] = \text{kg}, \quad [\text{vol}[\Omega]] = \text{m}^3.$$

Remark: Suppose that $\Omega_{\delta}(\vec{x})$ is a family of open sets all containing $\vec{x} \in B$ with the property that

$$\lim_{\delta \rightarrow 0} [\text{vol}(\Omega_{\delta})] = 0.$$

Then,

$$\rho(\vec{x}) = \lim_{\delta \rightarrow 0} \frac{\text{mass}[\Omega_{\delta}(\vec{x})]}{\text{vol}[\Omega_{\delta}(\vec{x})]}.$$

In fact, we can define the mass density by assuming that this limit exists for every $\vec{x} \in B$. //

Definition: Let B be a regular domain and $\rho: B \rightarrow \mathbb{R}$ be the mass density of B . The center of mass of $\Omega \subseteq B$ is

$$\vec{x}_{\text{com}}[\Omega] := \frac{1}{\text{mass}[\Omega]} \int_{\Omega} \vec{x} \rho(\vec{x}) dV_{\mathbf{x}}$$

The center of volume of $\Omega \subseteq B$ is

$$\vec{x}_{\text{cov}}[\Omega] = \frac{1}{\text{Vol}[\Omega]} \int_{\Omega} \vec{x} dV_{\vec{x}}.$$

Remark: It is possible that $\vec{x}_{\text{com}}[\Omega] \notin \Omega$
and $\vec{x}_{\text{cov}}[\Omega] \notin \Omega$. In general,

$$\vec{x}_{\text{com}}[\Omega] \neq \vec{x}_{\text{cov}}[\Omega].$$

But if $\rho(\vec{x}) \equiv \rho_0$, $\forall \vec{x} \in \Omega$, then

$$\vec{x}_{\text{com}}[\Omega] = \vec{x}_{\text{cov}}[\Omega].$$

Forces and Body Forces

Definition: A smooth vector field $\hat{\vec{b}}: B \rightarrow V$ is called a body force on B if it has units of force per unit volume, i.e.,

$$[\hat{\vec{b}}] = \frac{\text{Force}}{\text{Vol}} = \frac{N}{m^3}.$$

Let $\Omega \subseteq B$. The resultant force on Ω is

$$\vec{r}_b[\Omega] := \int_{\Omega} \vec{b}(\vec{x}) dV_{\vec{x}}$$

and has units of force, i.e.,

$$[\vec{r}_b[\Omega]] = N.$$

The resultant torque on Ω about $\vec{z} \in B$ is

$$\vec{T}_b[\Omega] = \int_{\Omega} (\vec{x} - \vec{z}) \times \hat{\vec{b}}(\vec{x}) dV_x.$$

The units of torque are

$$[\vec{T}_b[\Omega]] = \text{N} \cdot \text{m}. //$$

Definition: Let $\hat{\vec{b}}: B \rightarrow \mathcal{V}$ the body force field on B as defined above. The body force per unit mass (or specific body force) is defined as

$$\vec{b}(\vec{x}) := \frac{\hat{\vec{b}}(\vec{x})}{\rho(\vec{x})}.$$

Thus

$$\vec{T}_b[\Omega] = \int_{\Omega} \rho(\vec{x}) \vec{b}(\vec{x}) dV_{\vec{x}}$$

and

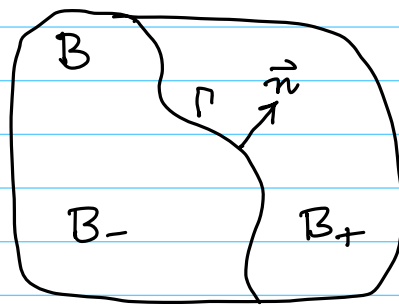
$$\vec{T}_b[\Omega] = \int_{\Omega} (\vec{x} - \vec{z}) \times \rho(\vec{x}) \vec{b}(\vec{x}) dV_{\vec{x}}$$

Body forces arise because of external influence on a body not because of physical contact.

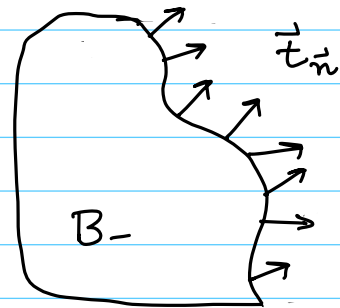
Surface forces, which we introduce next are forces induced by surface contact.

Surface Forces

We will differentiate between two types of surface forces, internal, along imaginary surfaces inside the body, and external, along the bounding surface of the body.



(a)



(b)

Let Γ be an internal (imaginary) bounding surface internal to the body B .

Suppose that Γ is orientable with normal $\vec{n}$ that points away from B_- .

At each point $\vec{x} \in \Gamma$ $\vec{n}: \Gamma \rightarrow \mathcal{V}$ defines a positive side and a negative side.

The force per unit area exerted by material on the positive side upon material on the negative side is assumed to be given by a function

$$\vec{t}_{\vec{n}}: \Gamma \rightarrow \mathcal{V},$$

which we call the traction or surface force field for Γ .

When Γ is the bounding surface of B , i.e., $\Gamma = \partial B$, we choose $\vec{n}$ as the outward pointing unit normal.

In this case $\vec{t}_{\vec{n}}: \partial B \rightarrow \mathcal{V}$ is the force per unit area exerted by the external body on the body B .

Of course

$$[\vec{t}_{\vec{n}}] = \frac{N}{m^2}$$

Definition: let Γ be an oriented surface in B and $\vec{t}_{\vec{n}}: \Gamma \rightarrow \mathcal{V}$ be the surface traction for Γ . The resulting force of the traction is

$$\vec{r}_s[\Gamma] := \int_{\Gamma} \vec{t}_{\vec{n}}(\vec{x}) dA_{\vec{x}}$$

The resulting torque about $\vec{z}$ due to the traction is defined as

$$\tau_s[\Gamma] := \int_{\Gamma} (\vec{x} - \vec{z}) \times \vec{t}_{\vec{n}}(\vec{x}) dA_{\vec{x}}$$

Cauchy's Postulate

Axiom (3.1): The traction field $\vec{t}_{\vec{n}}$ on a surface Γ in B depends only pointwise on the unit normal field

$$\vec{n}: \Gamma \rightarrow \mathcal{V}$$

In particular, there is a function

$$\vec{t}: \mathcal{N} \times \mathcal{B} \rightarrow \mathcal{V},$$

where $\mathcal{N} \subset \mathcal{V}$ is the set of unit vectors in $\mathcal{V}$, such that

$$\vec{t}_{\vec{n}}(\vec{x}) = \vec{t}(\vec{n}(\vec{x}), \vec{x}).$$

The function $t: N \times B \rightarrow V$ is called the traction function for B . ///

Result (3.2) let $\vec{t}: N \times B \rightarrow V$ be the traction function for a body B . Suppose that $t(\vec{n}, \vec{x})$ is continuous and that for any sequence Ω whose volumes tend to zero

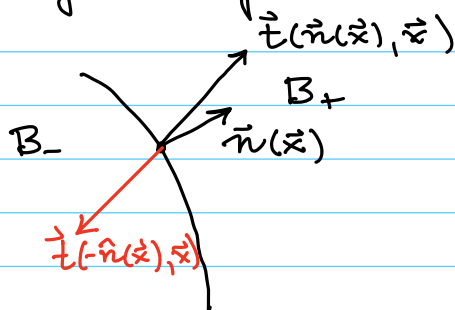
$$(3.1) \quad \begin{cases} \frac{1}{\text{area}[\partial\Omega]} \int_{\partial\Omega} t(\vec{n}(\vec{x}), \vec{x}) dA_{\vec{x}} \rightarrow 0 \\ \text{as } \text{vol}[\Omega] \rightarrow 0. \end{cases}$$

Then

$$t(-\vec{n}, \vec{x}) = -t(\vec{n}, \vec{x}), \quad \forall \vec{n} \in N, \forall \vec{x} \in B.$$

Proof: Book, page 81. ///

Remark: The result says that the traction exerted by material of the positive side on the negative side is equal and opposite to the traction exerted by the material of the negative side on the positive side.



The Stress Tensor

Result (3.3) (Cauchy's Theorem): Let $\vec{t}: N \times B \rightarrow V$ be the traction function for a body B and suppose it satisfies the conditions of Result 3.2. Then $\vec{t}(\vec{n}, \vec{x})$ is linear in $\vec{n}$, that is, for each $\vec{x} \in B$ there is a second-order tensor $S(\vec{x}) \in V^2$ such that

$$\vec{t}(\vec{n}, \vec{x}) = S(\vec{x})\vec{n}.$$

The field $S: B \rightarrow V^2$ is called the Cauchy Stress field for B .

Proof: Book, pages 83 and 84. //

let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. The components of the stress tensor can be found via

$$S_{ij}(\vec{x}) = \vec{e}_i \cdot (S(\vec{x})\vec{e}_j)$$

Then,

$$S(\vec{x}) = S_{ij}(\vec{x}) \vec{e}_i \otimes \vec{e}_j.$$

Suppose that

$$\vec{t}(\vec{n}(\vec{x}), \vec{x}) = t_i(\vec{n}(\vec{x}), \vec{x}) \vec{e}_i, \quad \vec{n}(\vec{x}) = n_i(\vec{x}) \vec{e}_i$$

Then

$$t_i(\vec{n}(\vec{x}), \vec{x}) = S_{ij}(\vec{x}) n_j(\vec{x}).$$

Now

$$t_i(\vec{e}_k, \vec{x}) = S_{ij}(\vec{x}) \delta_{jk} \\ = S_{ik}(\vec{x}).$$

This gives us another way to view the components of the stress tensor

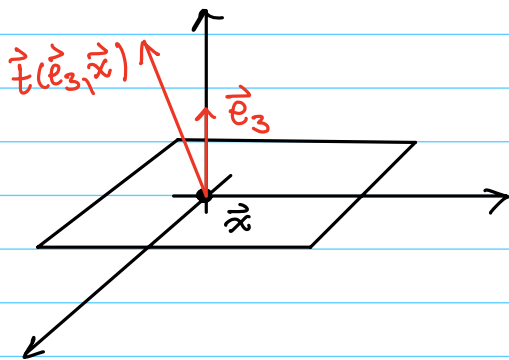
$$S_{ij}(\vec{x}) = t_i(\vec{e}_j, \vec{x}).$$

Thus,

$$\vec{t}(\vec{e}_1, \vec{x}) = t_i(\vec{e}_1, \vec{x}) \vec{e}_i = S_{i1}(\vec{x}) \vec{e}_i$$

$$\vec{t}(\vec{e}_2, \vec{x}) = t_i(\vec{e}_2, \vec{x}) \vec{e}_i = S_{i2}(\vec{x}) \vec{e}_i$$

$$\vec{t}(\vec{e}_3, \vec{x}) = t_i(\vec{e}_3, \vec{x}) \vec{e}_i = S_{i3}(\vec{x}) \vec{e}_i$$



Equilibrium

Assume that a body B has density $\rho: B \rightarrow \mathbb{R}$, applied external traction $\vec{h}: \partial B \rightarrow \mathcal{V}$ and applied body force per unit mass $\vec{b}: B \rightarrow \mathcal{V}$.

ρ , $\vec{h}$, and $\vec{b}$ are implicitly time independent.

We want to understand when B will be at equilibrium.

Axiom (3.4) (Conditions for Equilibrium): If a body in configuration B is in mechanical equilibrium, then the resultant force and resultant torque (about any fixed point $\vec{z}$) must vanish for every open subset $\Omega \subseteq B$. That is

$$\begin{aligned}
 \vec{F}[\Omega] &:= \vec{F}_b[\Omega] + \vec{F}_s[\partial\Omega] \\
 &= \int_{\Omega} \rho \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} \vec{t}(\vec{x}) dA_x = \vec{0}, \\
 \vec{T}[\Omega] &:= \vec{T}_b[\Omega] + \vec{T}_s[\partial\Omega] \\
 &= \int_{\Omega} (\vec{x} - \vec{z}) \times \rho(\vec{x}) \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} (\vec{x} - \vec{z}) \times \vec{t}(\vec{x}) dA_x = \vec{0}.
 \end{aligned}
 \tag{3.7}$$

Proposition: Suppose that

$$\vec{F}[\Omega] = \vec{0} \quad \forall \Omega \subseteq B.$$

Then, for all $\vec{z}$,

$$\begin{aligned}
 &\int_{\Omega} (\vec{x} - \vec{z}) \times \rho(\vec{x}) \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} (\vec{x} - \vec{z}) \times \vec{t}(\vec{x}) dA_x \\
 &= \int_{\Omega} \vec{x} \times \rho(\vec{x}) \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} \vec{x} \times \vec{t}(\vec{x}) dA_x.
 \end{aligned}$$

It follows that we can replace (3.7) by the equivalent form

$$(3.7) \quad \begin{aligned} \int_{\Omega} \rho \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} \vec{T}(\vec{x}) dA_x &= \vec{0}, \\ \int_{\Omega} \vec{x} \times \rho(\vec{x}) \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} \vec{x} \times \vec{T}(\vec{x}) dA_x &= \vec{0}. \end{aligned}$$

Local Equilibrium Conditions

Result (3.5): Assume that the Cauchy stress field S is cont. differentiable and the density field ρ and body force field $\vec{b}$ are continuous, then the equilibrium conditions in (3.7) are equivalent to

$$(3.8) \quad \begin{aligned} \nabla \cdot S(\vec{x}) + \rho(\vec{x}) \vec{b}(\vec{x}) &= \vec{0} \\ S^T(\vec{x}) &= S(\vec{x}) \end{aligned} \quad \forall \vec{x} \in B$$

Proof: ($\Rightarrow$) To establish (3.8), let us use

$$\vec{T}(\vec{x}) = S(\vec{x}) \vec{n}(x)$$

and (3.7), to get

$$\int_{\Omega} \rho \vec{b}(\vec{x}) dV_x + \int_{\partial\Omega} S(\vec{x}) \vec{n}(\vec{x}) dV_{\vec{x}} = \vec{0}$$

Using the Tensor Divergence Theorem (Result 2.17),

$$\int_{\partial\Omega} S \vec{n} dV_{\vec{x}} = \int_{\Omega} \nabla \cdot S dV_{\vec{x}}$$

for any $\Omega \subseteq B$. Thus

$$\int_{\Omega} (\nabla \cdot S + \rho \vec{b}) dV_{\vec{x}} = 0$$

By the localization Theorem (Result 2.19),

$$\nabla \cdot S + \rho \vec{b} = 0 \quad \text{in } B.$$

To establish (3.8)₂, again using the Cauchy stress tensor,

$$\int_{\partial\Omega} \vec{x} \times (S \vec{n}) dA_{\vec{x}} + \int_{\Omega} \vec{x} \times \rho \vec{b} dV_{\vec{x}} = 0$$

Since

$$\rho \vec{b} = -\nabla \cdot S$$

$$(3.9) \quad \int_{\partial\Omega} \vec{x} \times (S \vec{n}) dA - \int_{\Omega} \vec{x} \times (\nabla \cdot S) dV_{\vec{x}} = 0$$

Define

$$R_{il} := \epsilon_{ijk} x_j S_{kl}$$

$$R = R_{il} \vec{e}_i \otimes \vec{e}_l.$$

Then

$$R \vec{n} = \vec{x} \times S \vec{n} \quad (\text{check this})$$

Hence

$$(3.10) \quad \int_{\partial\Omega} \vec{R} \, dA_x - \int_{\Omega} \vec{x} \times (\nabla \cdot \vec{S}) \, dV_x = \vec{0}$$

The divergence Theorem implies

$$\int_{\Omega} \nabla \cdot \vec{R} \, dV_x - \int_{\Omega} \vec{x} \times \nabla \cdot \vec{S} \, dV_x = \vec{0}$$

Since $\Omega \subseteq B$ is arbitrary, the localization Theorem implies

$$\nabla \cdot \vec{R} - \vec{x} \times (\nabla \cdot \vec{S}) = \vec{0} \quad \text{in } B.$$

In component form,

$$(\epsilon_{ijk} x_j S_{kl})_{,l} - \epsilon_{ijk} x_j S_{kl,l} = 0 \quad \text{in } B.$$

The first term may be simplified as

$$\begin{aligned} (\epsilon_{ijk} x_j S_{kl})_{,l} &= \frac{\partial}{\partial x_l} (\epsilon_{ijk} x_j S_{kl}) \\ &= \epsilon_{ijk} \delta_{jl} S_{kl} + \epsilon_{ijk} x_j S_{kl,l} \\ &= \epsilon_{ijk} S_{kj} + \epsilon_{ijk} x_j S_{kl,l} \end{aligned}$$

Thus

$$\begin{aligned} 0 &= (\epsilon_{ijk} x_j S_{kl})_{,l} - \epsilon_{ijk} x_j S_{kl,l} \\ &= \epsilon_{ijk} S_{kj} + \cancel{\epsilon_{ijk} x_j S_{kl,l}} - \cancel{\epsilon_{ijk} x_j S_{kl,l}} \\ &= \epsilon_{ijk} S_{kj} \end{aligned}$$

Therefore,

$$\begin{aligned} 0 &= \epsilon_{ikj} S_{jk} && (\text{why?}) \\ &= -\epsilon_{ijk} S_{jk}. \end{aligned}$$

Thus

$$\epsilon_{ijk} [S_{kj} - S_{jk}] = 0$$

This implies, using $i=1, j=2, k=3$, that

$$\begin{aligned} 0 &= \epsilon_{123} [S_{32} - S_{23}] = S_{32} - S_{23} \\ &\Rightarrow S_{32} = S_{23}. \end{aligned}$$

In like manner, we can show

$$S_{ij} = S_{ji} \quad \text{for all } i, j \in \{1, 2, 3\}. //$$

($\Leftarrow$): This direction is easier, and left as an exercise. ///